



SRI RAMAKRISHNA ENGINEERING COLLEGE

[Educational Service : SNR Sons Charitable Trust]

Vattamalaipalayam, N.G.G.O. Colony Post,
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Department of Artificial Intelligence and Data Science

SELF LEARNING ASSIGNMENT

AI-DRIVEN CYBER-PHYSICAL ROBOTICS

Nikitha M – 71812311038

III B. TECH AI&DS – VI SEM

Academic Year 2025-2026

20AD215 – CYBER PHYSICAL SYSTEMS

Course Coordinator

Mrs.M.Divya, AP(OG) /AI&DS



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Department of Artificial Intelligence and Data Science

ALM

20AD215 – CYBER PHYSICAL SYSTEMS Assessment

S.No.	Roll No &Name	Content (4)	Implementation (4)	Output (2)	Total (10)
1.	NIKITHA M – 71812311038				

Course Coordinator
Mrs.M.Divya, AP(OG) /AI&DS

MAPPING

PO's - Mapping

POs	Definition	Justification
PO1	Engineering Knowledge: Apply knowledge of mathematics, science, engineering fundamentals, and engineering specialization to solve complex engineering problems.	Applied concepts of Artificial Intelligence, robotics, and cyber-physical systems to understand intelligent robotic operations.
PO2	Problem Analysis: Identify, formulate, review research literature, and analyze complex engineering problems to reach substantiated conclusions for sustainable development.	Analyzed robotic decision-making and real-time control problems using AI techniques.
PO3	Design/Development of Solutions: Design solutions for complex engineering problems and design system components or processes that meet specific needs with appropriate consideration for public health and safety, and cultural, societal, and environmental considerations.	Designed AI-driven robotic systems for automation and real-time applications.
PO4	Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge, including the design of experiments, modeling, analysis, and interpretation of data to provide valid conclusions.	NIL
PO5	Engineering Tool Usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modeling, while recognizing their limitations, to solve complex engineering problems.	Used AI/ML tools, IoT, and robotic frameworks for system development.
PO6	The Engineer and the world: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.	NIL
PO7	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.	NIL
PO8	Individual and Collaborative Teamwork: Function effectively as an individual and as a member or leader in diverse/multi-disciplinary teams	Worked collaboratively to analyze AI-driven robotics in CPS.
PO9	Communication: Communicate effectively and inclusively within the engineering community and society at large, by being able to comprehend and write effective reports and design documentation, and make effective presentations considering cultural, language, and learning differences.	Presented concepts through reports and documentation.

PO10	Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.	NIL
PO11	Life-long Learning: Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.	NIL.

PSO's – Mapping

PSOs	Definition	Justification
PSO1	Analyze, design and build sustainable intelligent solutions to solve challenges imposed by industry and society.	Studied AI-driven robotic CPS systems for real-world automation and intelligent solutions.
PSO2	Demonstrate data analysis skills to achieve effective insights and decision making to solve real-life problems.	Analyzed sensor data and decision-making in robotic systems.
PSO3	Apply mathematical and statistical models to solve the computational tasks, and model real-world problems using appropriate AI / ML algorithms.	NIL

SDG's – Mapping

SDGs	Definition	Justification
SDG1	No Poverty	NIL
SDG2	Zero Hunger	NIL
SDG3	Good Health and Well-being	NIL
SDG4	Quality Education	NIL
SDG5	Gender Equality	NIL
SDG6	Clean Water and Sanitation	NIL
SDG7	Affordable and Clean Energy	NIL
SDG8	Decent Work and Economic Growth	NIL
SDG9	Industry, Innovation and Infrastructure	AI-driven robotics improves automation, innovation, and industrial efficiency.
SDG10	Reduced Inequalities	NIL
SDG11	Sustainable Cities and Communities	NIL
SDG12	Responsible Consumption and Production	NIL
SDG13	Climate Action	NIL
SDG14	Life Below Water	NIL
SDG15	Life on Land	NIL
SDG16	Peace, Justice and Strong Institutions	NIL
SDG17	Partnerships for the Goals	NIL

INTRODUCTION

Cyber-Physical Systems (CPS) are systems that integrate physical processes with computational intelligence and communication networks. When Artificial Intelligence (AI) is incorporated into these systems, they transform into intelligent robotic platforms capable of real-time decision-making, automation, and adaptive behavior. AI-Driven Cyber-Physical Robotics refers to robotic systems that interact with the physical environment using sensors and actuators while utilizing AI algorithms to analyze data and make decisions. These systems are widely applied in domains such as manufacturing, healthcare, transportation, and smart environments. Unlike traditional robots, AI-driven CPS robots can learn from data, predict future outcomes, and automatically adjust their actions based on changes in the environment, thereby improving efficiency and performance.

TOOLS AND TECHNOLOGIES USED

- Artificial Intelligence is used to enable intelligent decision-making in robotic systems by applying Machine Learning and Deep Learning techniques.
- Machine Learning models help the system learn patterns from data and improve performance over time.
- Sensors are used to collect real-time data such as temperature, motion, and images from the physical environment.
- Actuators convert the processed decisions into physical actions such as movement and control operations.
- Embedded systems like Arduino and Raspberry Pi are used to process data and control hardware components in real time.
- These embedded systems provide efficient hardware interfacing and low power consumption.
- Communication technologies such as Wi-Fi, Bluetooth, and MQTT enable data exchange between devices in the system.
- Internet of Things (IoT) helps in connecting multiple components and ensuring smooth communication.
- Robotics frameworks like ROS are used to design, simulate, and control robotic systems efficiently.
- ROS provides modular architecture, simulation tools, and hardware abstraction for easy development.

SYSTEM WORKFLOW

The AI-driven CPS robotic system works as follows:

1. **Data Sensing**
Sensors collect real-time data from the environment.
2. **Data Processing**
The collected data is processed using embedded systems or cloud platforms.

3. **AI Decision Making**

AI algorithms analyze the data and make intelligent decisions.

4. **Control Execution**

Commands are sent to actuators to perform actions.

5. **Feedback Loop**

The system continuously monitors and updates actions based on new data.

IMPLEMENTATION STEPS

Step 1 – System Design

- Identify the problem (e.g., obstacle detection).
- Select sensors, controllers, and actuators.

Step 2 – Data Collection

- Collect real-world data using sensors.
- This data is used to train AI models.

Step 3 – AI Model Development

- Develop ML/DL models for decision making.

Examples:

- CNN for image classification , ML for path prediction

Step 4 – Hardware Integration

- Connect AI model with Arduino/Raspberry Pi.
- Enable communication with robotic components.

Step 5 – Real-Time Execution

- Deploy the system for real-time tasks like:
 - Autonomous navigation
 - Object detection
 - Smart control

Step 6 – Feedback & Optimization

- Monitor performance continuously.

- Improve accuracy using new data.

ADVANTAGES

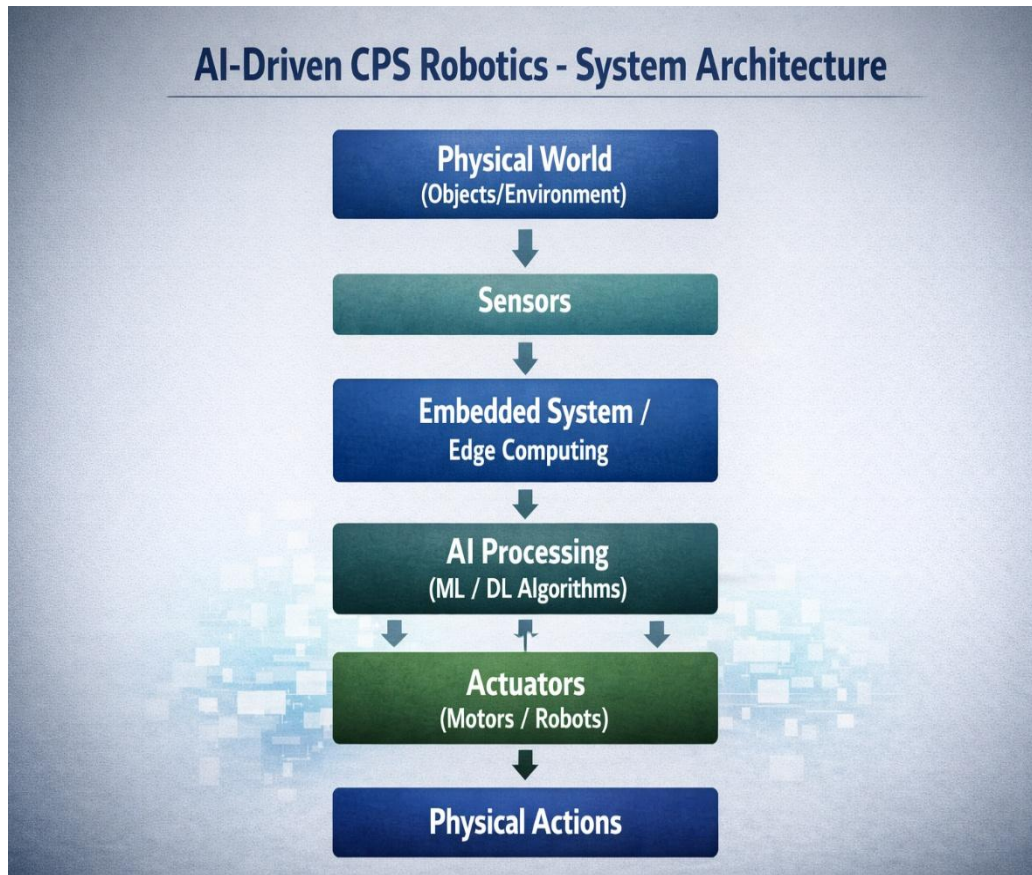
- AI-driven cyber-physical robotic systems enable real-time decision-making in dynamic environments.
- These systems reduce human effort by automating complex tasks.
- They improve efficiency and productivity in various applications.
- The integration of AI increases accuracy and system performance.
- These systems can learn from data and continuously improve over time.
- They are scalable and can be adapted to different use cases and environments.

LIMITATIONS

- The implementation cost of AI-driven CPS robotics is high due to advanced technologies.
- System integration is complex and requires knowledge from multiple domains.
- These systems are vulnerable to security risks due to network connectivity.
- The performance of the system depends on the quality of data used for training.
- Poor or insufficient data can lead to inaccurate results and decisions.

SYSTEM ARCHITECTURE

- The system begins with the physical world, which includes real-time environment and objects.
- Sensors collect data such as temperature, images, and motion from the environment.
- The collected data is sent to the embedded system or edge computing unit for initial processing.
- The processed data is forwarded to the AI processing unit.
- AI algorithms analyze the data and make intelligent decisions.
- Control signals are generated based on the AI output.
- These signals are sent to actuators such as motors and robotic systems.
- Actuators perform physical actions in the environment.
- The system continuously updates through a feedback loop for better performance.



RESULT

The AI-Driven Cyber-Physical Robotics system successfully integrates sensing, computation, and control to perform intelligent and autonomous operations. It demonstrates the ability to analyze real-time data using Artificial Intelligence algorithms and make accurate decisions accordingly. The system shows adaptability by continuously updating its actions based on feedback from the environment. It improves efficiency, accuracy, and overall system performance while reducing the need for human intervention. This approach proves to be highly effective in real-world applications such as industrial automation, healthcare systems, and smart environments, thereby highlighting the importance of integrating AI with cyber-physical robotic systems.

GEOTAG PHOTO

